

22050 Continuous-Time Signals and Linear Systems

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L09

Application of Laplace Transformation

- Differential equations**
- Circuit analysis**

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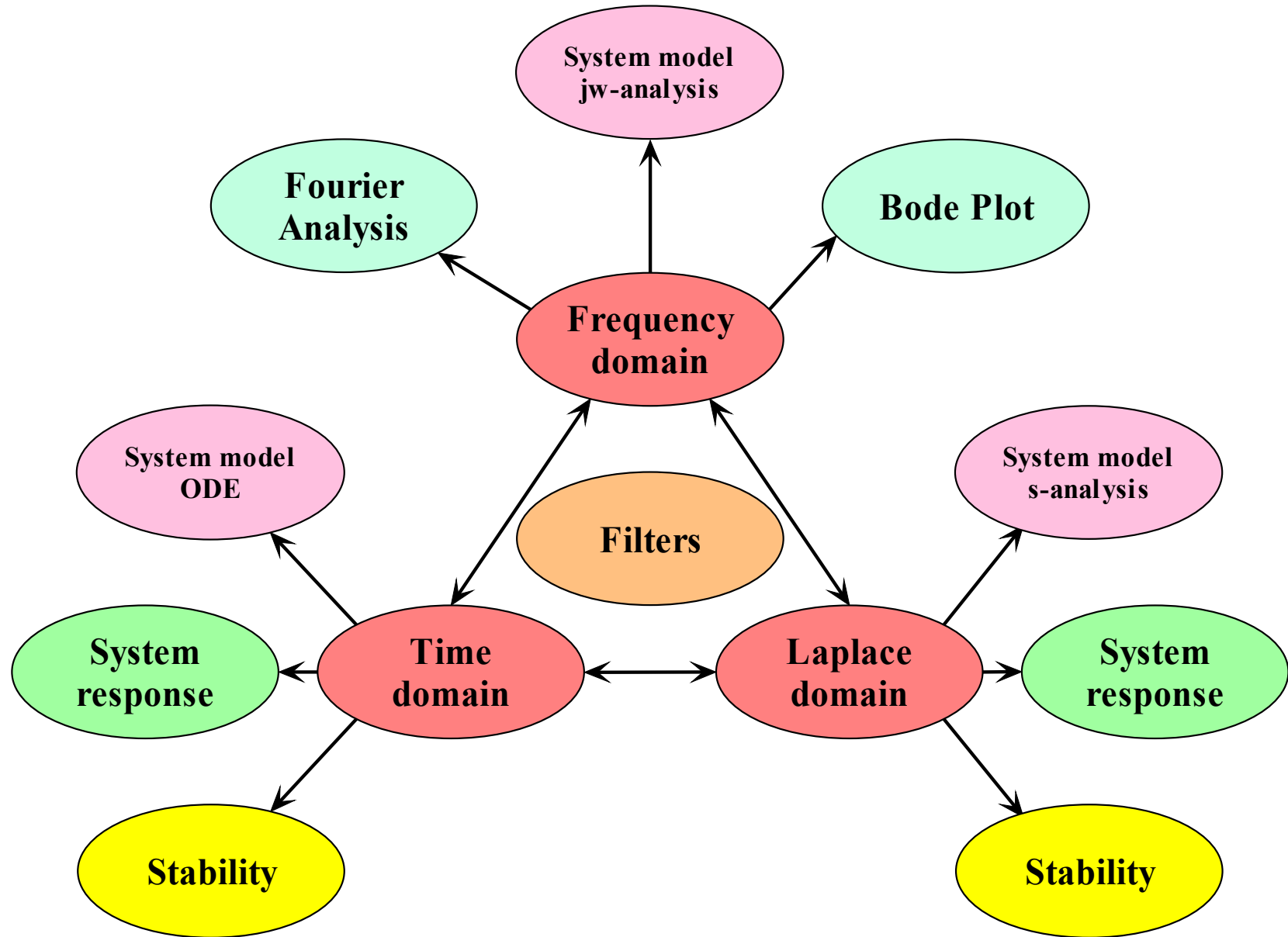
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6.3 Solutions to differential equation using Laplace transform

6.4 Analysis of electrical circuits using the Laplace transform

Time scaling:

$$x(at) \leftrightarrow \frac{1}{a} X\left(\frac{s}{a}\right), a > 0$$

Time shift:

$$x(t - t_0) \leftrightarrow e^{-st_0} X(s), t_0 \geq 0$$

Frequency shift theorem:

$$x(t)e^{s_0 t} \leftrightarrow X(s - s_0)$$

Convolution theorem:

$$x_1(t) * x_2(t) \leftrightarrow X_1(s) \cdot X_2(s)$$

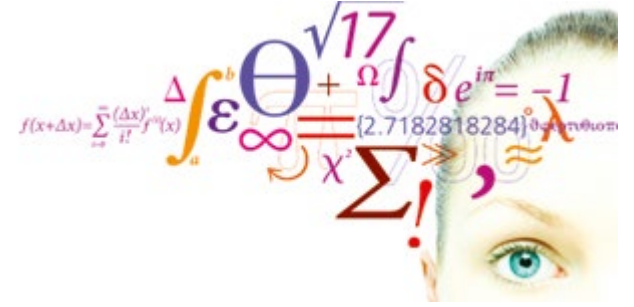
Differentiation theorem:

$$\frac{dx}{dt} \leftrightarrow sX(s) - x(0_-)$$

$$\frac{d^2x}{dt^2} \leftrightarrow s^2X(s) - sx(0_-) - \dot{x}(0_-)$$

Integration theorem:

$$\int_{0_-}^t x(\tau) d\tau \leftrightarrow \frac{X(s)}{s} \qquad \int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{X(s)}{s} + \frac{\int_{-\infty}^{0_-} x(\tau) d\tau}{s}$$



Lathi 6.3

Solving differential equations using Laplace transformation

Video 1

Solve for the zero-input response and the zero-state response and add them together to get the total solution.

Transform pairs:

$$\ddot{y} + 4\dot{y} + 4y = 2\ddot{x}, \quad y(0_-) = 1, \dot{y}(0_-) = -1$$

$$x(t) = u(t)$$

$$y(t) \leftrightarrow Y(s) \quad \text{and} \quad \dot{y}(t) \leftrightarrow sY(s) - y(0_-)$$

$$\ddot{y}(t) \leftrightarrow s^2Y(s) - sy(0_-) - \dot{y}(0_-)$$

$$\ddot{x}(t) \leftrightarrow s^2X(s) \quad \text{and} \quad \dot{x}(0_-) = \ddot{x}(0_-) = 0 : \text{causal signal}$$

$$[s^2 + 4s + 4]Y(s) = 2s^2X(s) + \underbrace{sy(0_-) + \dot{y}(0_-) + 4y(0_-)}_{\text{initial condition terms}}$$

$$Y(s) = \frac{2s^2}{s^2 + 4s + 4} \cdot X(s) + \frac{sy(0_-) + \dot{y}(0_-) + 4y(0_-)}{s^2 + 4s + 4}$$

Generic form:

$$Y(s) = \underbrace{\frac{P(s)}{Q(s)}}_{\text{zero-state}} \cdot X(s) + \underbrace{\frac{I(s)}{Q(s)}}_{\text{zero-input}}$$

Zero-input response using Laplace Transformation

$$Y_{zi}(s) = \frac{I(s)}{Q(s)} = \frac{sy(0_-) + \dot{y}(0_-) + 4y(0_-)}{s^2 + 4s + 4} = \frac{s + 4 - 1}{s^2 + 4s + 4} = \frac{s + 3}{(s + 2)^2}$$

Before going any further we will use the initial and final value theorems to predict $y_{zi}(0_+)$ and $y_{zi}(t \rightarrow \infty)$.

$$y_{zi}(0_+) = \lim_{s \rightarrow \infty} s Y_{zi}(s) = \lim_{s \rightarrow \infty} s \frac{s + 3}{(s + 2)^2} = \frac{s^2}{s^2} = 1$$

$$\lim_{t \rightarrow \infty} y_{zi}(t) = \lim_{s \rightarrow 0} s Y_{zi}(s) = \lim_{s \rightarrow 0} s \frac{s + 3}{(s + 2)^2} = \frac{0}{4} = 0$$

Zero-input response

using Laplace
Transformation

Partial fraction expansion

$$Y_{zi}(s) = \frac{s+3}{(s+2)^2} = \frac{k_1}{(s+2)^2} + \frac{k_2}{s+2}$$

$$k_1 = (s+2)^2 \frac{(s+3)}{(s+2)^2} \Big|_{s=-2} - \cancel{(s+2)^2 \frac{k_2}{s+2} \Big|_{s=-2}} = (s+3) \Big|_{s=-2} = 1$$

Pick an arbitrary value
for s not used before.

$s = 0$

$$\frac{s+3}{(s+2)^2} = \frac{1}{(s+2)^2} + \frac{k_2}{s+2} \quad \frac{3}{4} = \frac{1}{4} + \frac{k_2}{2} \Rightarrow k_2 = 1$$

$$Y_{zi}(s) = \frac{s+3}{(s+2)^2} = \frac{1}{(s+2)^2} + \frac{1}{s+2}$$

Zero-input response:

$$y_{zi}(t) = te^{-2t}u(t) + e^{-2t}u(t) = (1+t)e^{-2t}u(t)$$

Checking with our predictions:

$$y_{zi}(0_+) = 1 \quad \lim_{t \rightarrow \infty} y_{zi}(t) = 0$$

Zero-state response using Laplace Transformation:

$$Y_{zs}(s) = \frac{P(s)}{Q(s)} \cdot X(s) = H(s)X(s) \quad x(t) = u(t) \leftrightarrow X(s) = \frac{1}{s}$$

$$Y_{zs}(s) = \frac{2s^2}{(s+2)^2} \cdot \frac{1}{s} = \frac{2s}{(s+2)^2}$$

Before going any further we will use the initial and final value theorems to predict $y_{zs}(0_+)$ and $y_{zs}(t \rightarrow \infty)$.

$$y_{zs}(0_+) = \lim_{s \rightarrow \infty} s Y_{zs}(s) = \lim_{s \rightarrow \infty} s \frac{2s}{(s+2)^2} = \frac{2s^2}{s^2} = 2$$

$$\lim_{t \rightarrow \infty} y_{zs}(t) = \lim_{s \rightarrow 0} s Y_{zs}(s) = \lim_{s \rightarrow 0} s \frac{2s}{(s+2)^2} = \frac{0}{4} = 0$$

Solving Differential Equations using Laplace Transformation

Partial fraction expansion

$$Y_{zs}(s) = \frac{2s^2}{(s+2)^2} \cdot \frac{1}{s} = \frac{2s}{(s+2)^2} = \frac{k_1}{(s+2)^2} + \frac{k_2}{s+2}$$

$$k_1 = 2s \Big|_{s=-2} = -4 \qquad k_2: s \stackrel{\text{def}}{=} 0 \Rightarrow 0 = \frac{-4}{4} + \frac{k_2}{2} \Rightarrow k_2 = 2$$

$$Y_{zs}(s) = \frac{2s}{(s+2)^2} = \frac{-4}{(s+2)^2} + \frac{2}{s+2}$$

$$y_{zs}(t) = -4te^{-2t}u(t) + 2e^{-2t}u(t)$$

Zero-state response:

$$y_{zs}(t) = (2 - 4t)e^{-2t}u(t)$$

Checking with our predictions:

$$y_{zs}(0_+) = 2 \qquad \lim_{t \rightarrow \infty} y_{zs}(t) = 0$$

Zero-input response:

$$y_{zi}(t) = (1 + t)e^{-2t}u(t)$$

Zero-state response:

$$y_{zs}(t) = (2 - 4t)e^{-2t}u(t)$$

Using the "Signals and Linear Systems approach", we obtain separate expressions for:

- Zero-input response
- Zero-state response

This way we can state precisely how the input signal contributes to the output signal.

$$\begin{aligned} y(t) &= y_{zi}(t) + y_{zs}(t) \\ &= \underbrace{(1 + t)e^{-2t}u(t)}_{\text{zero-input}} + \underbrace{(2 - 4t)e^{-2t}u(t)}_{\text{zero-state}} \end{aligned}$$

If we had used the classical approach where the initial conditions are given at 0_+ , then we would not be able to distinguish the two responses.

$$y(t) = (3 - 3t)e^{-2t}u(t)$$

Example:

$$H(s) = \frac{2s^2}{(s+2)^2}$$

This example is a challenge to partial fraction expansion.

We need to simplify the expression by using polynomial division.

Polynomial division

$$\begin{array}{r} 2 \\ (s+2)^2 \overline{) 2s^2} \\ \underline{2s^2 + 8s + 8} \\ -8s - 8 \end{array}$$

$$\frac{2s^2}{(s+2)^2} = 2 - 8 \frac{s+1}{(s+2)^2}$$

Now we have two terms. We only need to do partial fraction expansion on the latter.

$$H(s) = \frac{2s^2}{(s+2)^2} = 2 - 8 \frac{s+1}{(s+2)^2}$$

$$H(s) = \frac{2s^2}{(s+2)^2} = 2 - 8 \frac{s+1}{(s+2)^2}$$

Predicting initial value:

$$\lim_{s \rightarrow \infty} 8s \frac{s+1}{(s+2)^2} = -8 \lim_{s \rightarrow \infty} \frac{s^2}{s^2} = -8$$

If the limit on the right-hand side exists.

Predicting final value:

$$\lim_{t \rightarrow \infty} h(t) = \lim_{s \rightarrow 0} s H(s) = \lim_{s \rightarrow 0} 2s - \lim_{s \rightarrow 0} 8s \frac{s+1}{(s+2)^2} = 0$$

Partial fraction expansion:

$$H(s) = \frac{2s^2}{(s+2)^2} = 2 - 8 \frac{s+1}{(s+2)^2}$$

$$= 2 + \frac{8}{(s+2)^2} - \frac{8}{s+2}$$

Looking up the inverse transforms in table 6.1:

$$h(t) = 2\delta(t) + 8te^{-2t}u(t) - 8e^{-2t}u(t)$$

$$h(t) = 2\delta(t) + 8(t-1)e^{-2t}u(t)$$

Checking with our predictions:

$$h(0_+) = -8 \quad \lim_{t \rightarrow \infty} h(t) = 0$$

Observe:

The power of s in the numerator is equal to the power of s in the denominator. A clear sign that this is a **high-pass** filter. And we got the $\delta(t)$ function we expected.

The initial value theorem predicts the value at 0_+ . At $t = 0_+$ the impulse has vanished.

Another example.

Solve for the zero-input response and the zero-state response and add them together to get the total solution.

Transform pairs:

$$\ddot{y} + 4\dot{y} + 3y = 2\dot{x} + x, \quad y(0_-) = 1, \dot{y}(0_-) = 2$$

$$x(t) = u(t)$$

$$y(t) \leftrightarrow Y(s) \quad \text{and} \quad \dot{y}(t) \leftrightarrow sY(s) - y(0_-)$$

$$\ddot{y}(t) \leftrightarrow s^2Y(s) - sy(0_-) - \dot{y}(0_-)$$

$$\dot{x}(t) \leftrightarrow sX(s) \quad \text{and} \quad \dot{x}(0_-) = \ddot{x}(0_-) = 0 : \text{causal signal}$$

$$[s^2 + 4s + 3]Y(s) = (2s + 1)X(s) + sy(0_-) + \dot{y}(0_-) + 4y(0_-)$$

$$Y(s) = \frac{2s + 1}{s^2 + 4s + 3} \cdot X(s) + \frac{sy(0_-) + \dot{y}(0_-) + 4y(0_-)}{s^2 + 4s + 3}$$

Generic form:

$$Y(s) = \underbrace{\frac{P(s)}{Q(s)}}_{\text{zero-state}} \cdot X(s) + \underbrace{\frac{I(s)}{Q(s)}}_{\text{zero-input}}$$

Before going any further we will use the initial and final value theorems to predict $y_{zi}(0_+)$ and $y_{zi}(t \rightarrow \infty)$.

$$Y_{zi}(s) = \frac{I(s)}{Q(s)} = \frac{s + 6}{s^2 + 4s + 3}$$

Initial value:
$$y_{zi}(0_+) = \lim_{s \rightarrow \infty} s Y_{zi}(s) = \lim_{s \rightarrow \infty} s \frac{s + 6}{s^2 + 4s + 3} = \frac{s^2}{s^2} = 1$$

Final value:
$$\lim_{t \rightarrow \infty} y_{zi}(t) = \lim_{s \rightarrow 0} s Y_{zi}(s) = \lim_{s \rightarrow 0} s \frac{s + 6}{s^2 + 4s + 3} = \frac{0}{3} = 0$$

Zero-input response:

$$Y_{zi}(s) = \frac{I(s)}{Q(s)} = \frac{s+6}{s^2+4s+3} = \frac{s+6}{(s+1)(s+3)} = \frac{k_1}{s+1} + \frac{k_2}{s+3}$$

Partial fraction expansion:

$$k_1 = \left. \frac{(s+6)}{(s+3)} \right|_{s=-1} = \frac{5}{2} \qquad k_2 = \left. \frac{(s+6)}{(s+1)} \right|_{s=-3} = -\frac{3}{2}$$

$$Y_{zi}(s) = \frac{1}{2} \left(\frac{5}{s+1} - \frac{3}{s+3} \right)$$

Inverse Laplace transform:

$$y_{zi}(t) = \left(\frac{5}{2} e^{-t} - \frac{3}{2} e^{-3t} \right) u(t)$$

Zero-state response:

$$Y_{zs}(s) = \frac{P(s)}{Q(s)} \cdot X(s) = H(s)X(s) \quad x(t) = u(t) \leftrightarrow X(s) = \frac{1}{s}$$

Partial fraction expansion:

$$Y_{zs}(s) = \frac{2s + 1}{s^2 + 4s + 3} \cdot \frac{1}{s} = \frac{2s + 1}{s(s + 1)(s + 3)} = \frac{k_1}{s} + \frac{k_2}{s + 1} + \frac{k_3}{s + 3}$$

$$k_1 = \left. \frac{2s + 1}{(s + 1)(s + 3)} \right|_{s=0} = \frac{1}{3} \quad k_2 = \left. \frac{2s + 1}{s(s + 3)} \right|_{s=-1} = \frac{1}{2} \quad k_3 = \left. \frac{2s + 1}{s(s + 1)} \right|_{s=-3} = \frac{-5}{6}$$

$$Y_{zs}(s) = \frac{1/3}{s} + \frac{1/2}{s + 1} - \frac{5/6}{s + 3}$$

Inverse Laplace transform:

$$y_{zs}(t) = \left(\frac{1}{3} + \frac{1}{2}e^{-t} - \frac{5}{6}e^{-3t} \right) u(t)$$

Total solution:

$$\begin{aligned}
 y(t) &= y_{zi}(t) + y_{zs}(t) \\
 &= \underbrace{\left(\frac{5}{2}e^{-t} - \frac{3}{2}e^{-3t} \right) u(t)}_{\text{zero-input}} + \underbrace{\left(\frac{1}{3} + \frac{1}{2}e^{-t} - \frac{5}{6}e^{-3t} \right) u(t)}_{\text{zero-state}} \\
 &= \underbrace{\left(\frac{1}{3} + 3e^{-t} - \frac{7}{3}e^{-3t} \right) u(t)}_{\text{classical}}
 \end{aligned}$$

Again, we see that the principle of splitting the total response into a zero-input and a zero-state response enables us to distinguish the impact of the input signal from the response to release of internal energy (zero-input response).

By Laplace transformation of differential equations, we have shown:

$$Q(D) y_{zs}(t) = P(D) x(t)$$

$$Q(s) Y(s) = P(s) X(s) + I(s)$$

$$Y_{zs}(s) = \frac{P(s)}{Q(s)} \cdot X(s)$$

We define the **transfer function** $H(s)$ as the zero-state response divided by the input signal:

The roots of the characteristic equation are the poles of the transfer function (only if common factors have not been cancelled out):

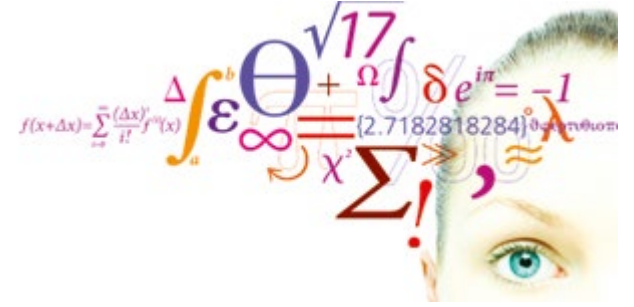
$$H(s) \stackrel{\text{def}}{=} \frac{Y_{zs}(s)}{X(s)} = \frac{P(s)}{Q(s)}$$

$$y_{zs}(t) = h(t) * x(t)$$

$$Y_{zs}(s) = H(s) \cdot X(s)$$

$$h(t) \leftrightarrow H(s)$$

The transfer function is the Laplace transform of the impulse response:



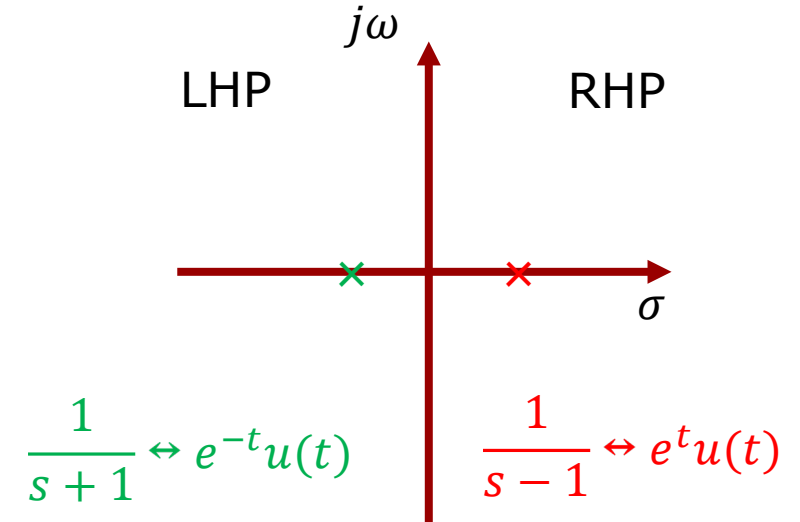
Lathi 6.3

Stability analysis using Laplace transformation

We can now state the conditions of stability in terms of the poles of the transfer function.

1. An LTIC system is **asymptotically stable** if and only if all the poles of the transfer function $H(s)$ are in the left-half-plane (**LHP**). Poles in the LHP may be repeated or unrepeated.
2. An LTIC system is **unstable** if and only if either one or both of the following conditions are met: (i) at least one pole of $H(s)$ is in the **RHP**, (ii) there are repeated poles of $H(s)$ on the imaginary axis.
3. An LTIC system is **marginally stable** if and only if there are no poles of $H(s)$ in the RHP, and there are some *unrepeated* poles on the imaginary axis.

$$\frac{1}{s+a} \leftrightarrow e^{-at}u(t)$$



Direct quote from Lathi and Green, p. 555.

Observation:

A zero may cancel a *pole* in the transfer function.

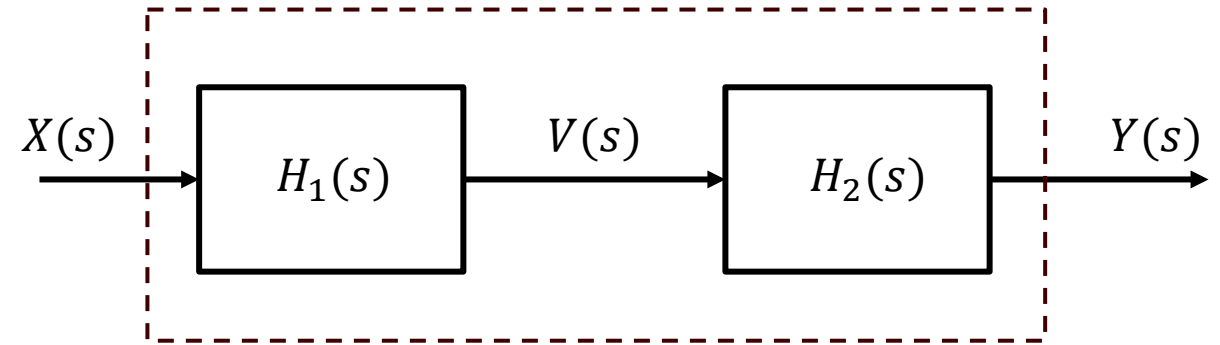
However, zeros have no impact on the zero-input response. A pole in the RHP will thus give an unbounded response to a non-zero initial state, even if there is a superimposed zero.

$$Y(s) = \frac{P(s)}{Q(s)} X(s) + \frac{I(s)}{Q(s)}$$

$$Y(s) = \underbrace{\frac{(s+1)\cancel{(s-2)}}{(s+2)\cancel{(s-2)}}}_{\text{zero-state response}} X(s) + \underbrace{\frac{I(s)}{(s+2)\cancel{(s-2)}}}_{\text{zero input response}}$$

$$Y(s) = \underbrace{\frac{(s+1)}{(s+2)}}_{\text{stable}} X(s) + \underbrace{\frac{I(s)}{(s+2)\cancel{(s-2)}}}_{\text{unstable}}$$

Here are shown a system, which, seen from the outside, appears to be one system, but in fact is composed of two subsystems in series.



$$V(s) = \frac{P_1(s)}{Q_1(s)} X(s) + \frac{I_1(s)}{Q_1(s)}$$

$$Y(s) = \frac{P_2(s)}{Q_2(s)} V(s) + \frac{I_2(s)}{Q_2(s)}$$

$$Y(s) = \frac{P_2(s)}{Q_2(s)} \left(\frac{P_1(s)}{Q_1(s)} X(s) + \frac{I_1(s)}{Q_1(s)} \right) + \frac{I_2(s)}{Q_2(s)}$$

$$Y(s) = \left(\frac{P_2(s)}{Q_2(s)} \frac{P_1(s)}{Q_1(s)} X(s) + \frac{P_2(s)}{Q_2(s)} \frac{I_1(s)}{Q_1(s)} \right) + \frac{I_2(s)}{Q_2(s)}$$

If we monitor the input and output signals, we may conclude that the signals are bounded in amplitude and hence that the system is stable in the Bounded Input - Bounded Output (BIBO) sense.

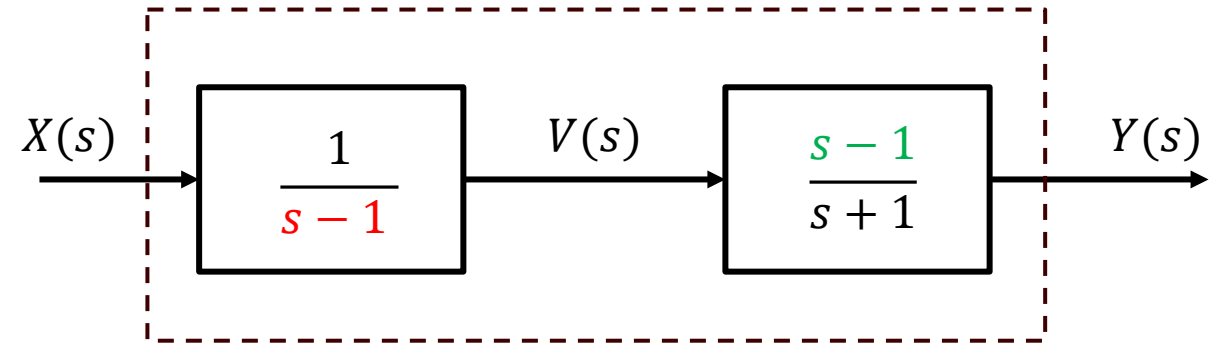
Let us look at an example.

Transfer function $H_1(s)$ has a pole ($s - 1$) in the right-half plane. Hence subsystem 1 is unstable. But what about the composite system?

Subsystem 2 has a zero ($s - 1$) that attempts to cancel the troublesome pole of subsystem 1.

Doing the math, we end up with an equation which features only a pole ($s + 1$) in the left half plane. The impulse response of the composite system is $e^{-t}u(t)$, hence the composite system appears to be asymptotic stable (only poles in the half-plane).

However, things are much worse than they appear.



$$V(s) = \frac{1}{s-1} X(s) + \frac{I_1(s)}{s-1}$$

$$Y(s) = \frac{s-1}{s+1} V(s) + \frac{I_2(s)}{s+1}$$

$$Y(s) = \frac{s-1}{s+1} \left(\frac{1}{s-1} X(s) + \frac{I_1(s)}{s-1} \right) + \frac{I_2(s)}{s+1}$$

$$Y(s) = \left(\cancel{\frac{s-1}{s+1}} \frac{1}{\cancel{s-1}} X(s) + \cancel{\frac{s-1}{s+1}} \frac{I_1(s)}{\cancel{s-1}} \right) + \frac{I_2(s)}{s+1}$$

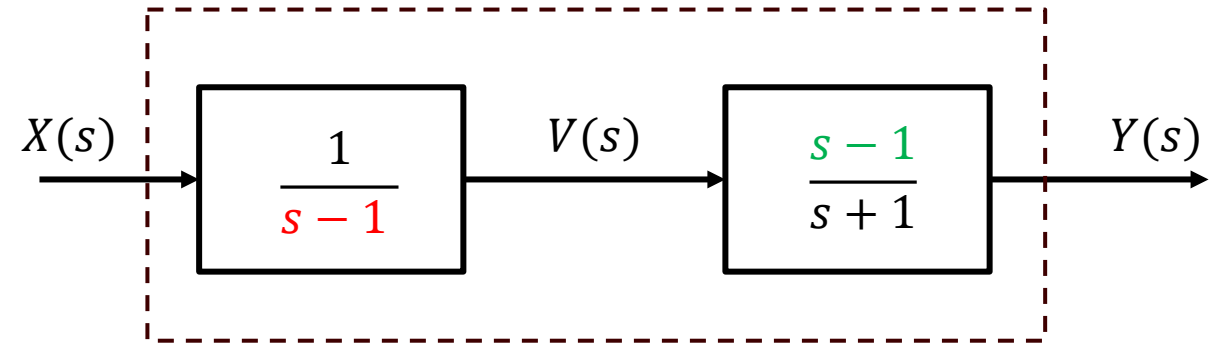
$$Y(s) = \left(\frac{1}{s+1} X(s) + \frac{I_1(s)}{s+1} \right) + \frac{I_2(s)}{s+1}$$

Transfer function $H_1(s)$ has a pole ($s - 1$) in the right-half plane. Hence subsystem 1 has an impulse response $e^t u(t)$. This behavior is independent of subsystem 2. Subsystem 2 cannot prevent the internal signal $v(t)$ from growing exponentially.

We notice also that $v(t)$ will grow exponentially whether stimulated by the input signal or simply from the presence of internal energy stored in subsystem 1.

In an electronic system, subsystem 1 would enter into a **mode of saturation**, thus clipping $v(t)$ at the supply voltage. Subsystem 2 cannot rectify this, but it will give the erroneous impression of a bounded output.

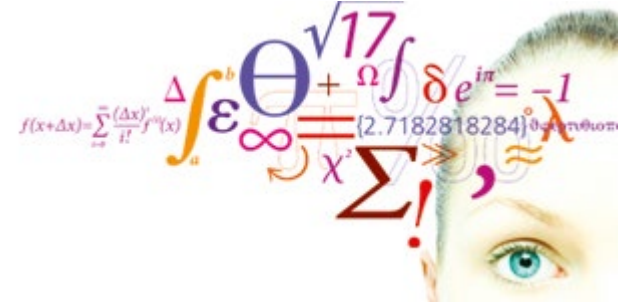
If subsystem 1 were a mechanical system, it would likely self-destruct.



$$V(s) = \frac{1}{s - 1} X(s) + \frac{I_1(s)}{s - 1}$$

If a system $H_1(s)$ has an undesired performance, adding a compensating system $H_2(s)$ may in many cases improve performance.

Local internal instability cannot be compensated downstream.



Lathi 6.4

Analyzing electric circuits using Laplace transformation

Video 2

The capacitor:

The voltage drop over a capacitor is the sum of the initial voltage and the voltage due to the charging current.

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{X(s)}{s} + \frac{\int_{-\infty}^{0-} x(\tau) d\tau}{s}$$

$$v_C(t) = \frac{1}{C} \int_{0-}^t i_C(\tau) d\tau + v_C(0-)$$

$$V_C(s) = \frac{1}{sC} I_C(s) + \frac{v_C(0-)}{s}$$

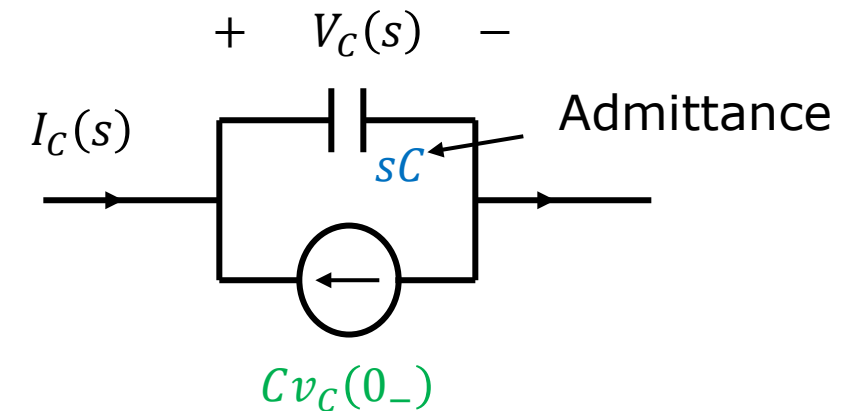
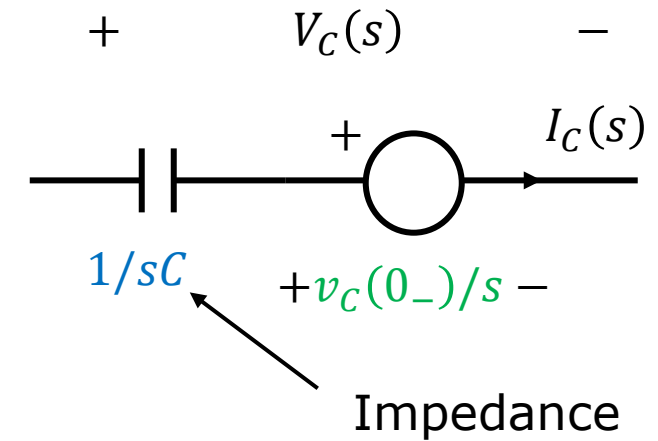
$$i_C(t) = C \frac{dv_C}{dt}$$

$$I_C(s) = sC V_C(s) - C v_C(0-)$$

Solving for the total current, we get a sum of two terms.

Hence, we obtain a model with two components in parallel.

We can think of the current source (green term) as a discharge current (or leakage current) reflecting the initial charged state of the capacitor. The blue term is the charging current needed to change the voltage at a given rate, if there were no initial charged state.



The inductor:

The voltage drop over an inductor is proportional to the rate of change in current.

Solving for the total current, we get a sum of two terms.

Hence, we obtain a model with two components in parallel.

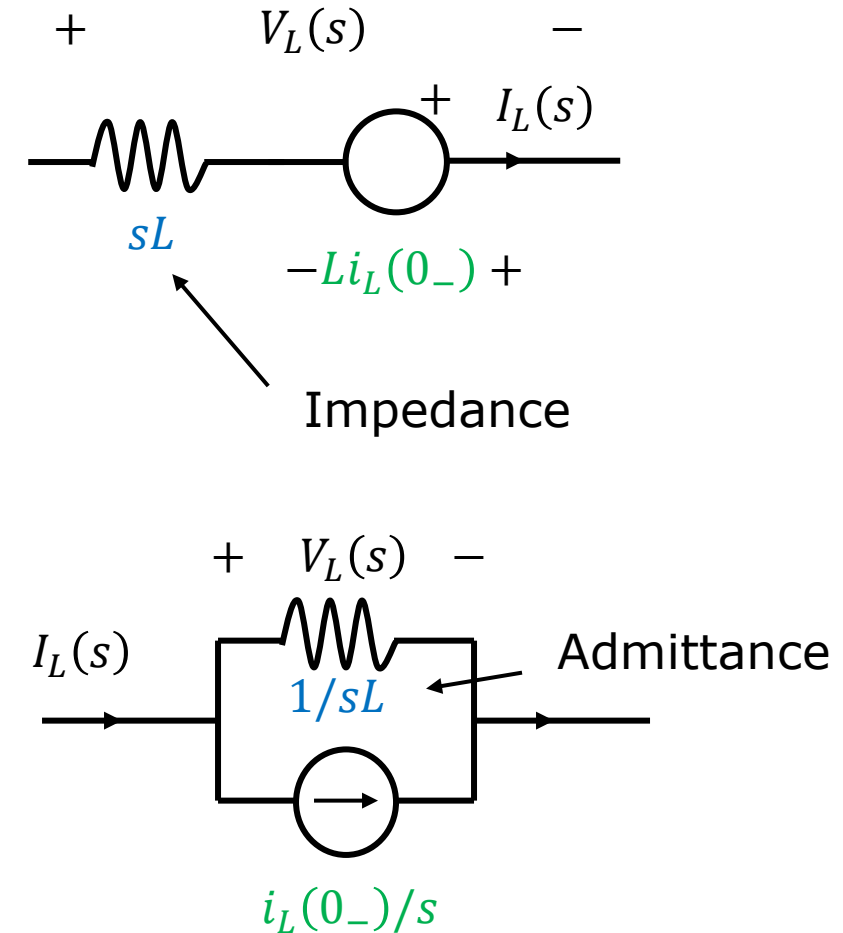
The source (green) term represents the initial magnetized state of the inductor. An increase in current increases magnetic energy. A release of stored magnetic energy induces a current. The green source represents the release of an initial magnetic energy stored in the inductor.

$$v_L(t) = L \frac{di_L}{dt}$$

$$V_L(s) = sL I_L(s) - L i_L(0_-)$$

$$i_L(t) = \frac{1}{L} \int_{0_-}^t v_L(\tau) d\tau + i_L(0_-)$$

$$I_L(s) = \frac{1}{sL} V_L(s) + \frac{i_L(0_-)}{s}$$



Observation:

For both the capacitor and the inductor we have two models (parallel or serial) to choose from. They are equally valid, but one may be more convenient to use than the other.

- **Nodal analysis:** Here we sum up currents. In this case the models with parallel components may be preferred because the two branches in the model are both currents.

$$I_C(s) = sC V_C(s) - C v_C(0_-)$$

$$I_L(s) = \frac{1}{sL} V_L(s) + \frac{i_L(0_-)}{s}$$

- **Mesh analysis:** Here we sum up voltage change in a loop. In this case the series model models may be preferred, because it adds voltages.

$$V_C(s) = \frac{1}{sC} I_C(s) + \frac{v_C(0_-)}{s}$$

$$V_L(s) = sL I_L(s) - L i_L(0_-)$$

Example 1.

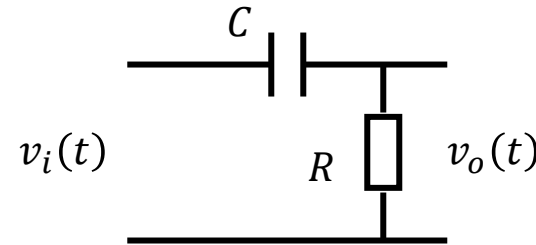
Laplace transform this circuit:

Using Kirchhoff's current law and Ohm's law for impedances in the s – domain:

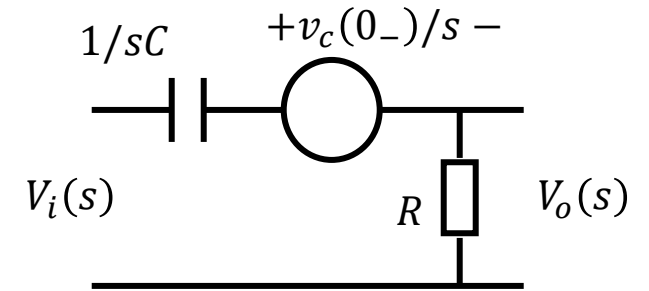
Simplifying and solving for $V_o(s)$:

We obtain a first order highpass filter.

Time domain



s domain



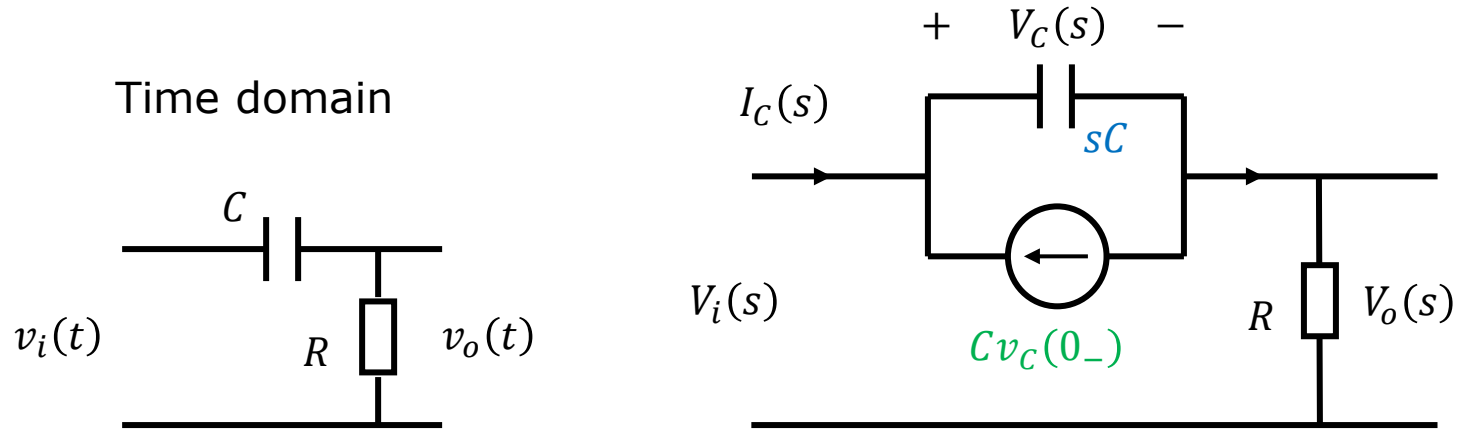
$$\frac{V_i(s) - \left(\frac{v_c(0_-)}{s} + V_o(s) \right)}{1/sC} = \frac{V_o(s)}{R}$$

The series model adds a voltage term.

$$sRCV_i(s) - RCv_c(0_-) = (sRC + 1)V_o(s)$$

$$V_o(s) = \underbrace{\frac{s}{s + 1/RC} V_i(s)}_{\text{zero-state}} - \underbrace{\frac{v_c(0_-)}{s + 1/RC}}_{\text{zero-input}}$$

Example 1 (alternative).



Using Kirchhoff's current law and Ohm's law for impedances in the s – domain:

$$I_c(s) = sCV_c(s) - Cv_c(0_-)$$

$$sCV_c(s) - Cv_c(0_-) = \frac{V_o(s)}{R}$$

$$sRC(V_i(s) - V_o(s)) - RCv_c(0_-) = V_o(s)$$

$$(sRC + 1)V_o(s) = sRCV_i(s) - RCv_c(0_-)$$

Simplifying and solving for $V_o(s)$:

We obtain a first order highpass filter.

$$V_o(s) = \underbrace{\frac{s}{s + 1/RC} V_i(s)}_{\text{zero-state}} - \underbrace{\frac{v_c(0_-)}{s + 1/RC}}_{\text{zero-input}}$$

Setting $V_i(s) = 2/s$
and $v_C(0_-) = 1$ V:

Before computing the inverse Laplace transforms of the two terms, we would like to use the **initial and final value theorems** to predict the initial and final values of the two terms in the time domain:

Question:

How is it possible that the output dies away in response to a step input?

$$V_o(s) = \underbrace{\frac{s}{s + 1/RC} \cdot \frac{2}{s}}_{\text{zero-state}} - \underbrace{\frac{1}{s + 1/RC}}_{\text{zero-input}}$$

Initial values: $v_{o,ZI}(0_+) = \lim_{s \rightarrow \infty} s \cdot \frac{-1}{s + 1/RC} = -1$

$$v_{o,ZS}(0_+) = \lim_{s \rightarrow \infty} s \cdot \frac{2}{s + 1/RC} = 2$$

Final values: $\lim_{t \rightarrow \infty} v_{o,ZI}(t) = \lim_{s \rightarrow 0} s \cdot \frac{-1}{s + 1/RC} = 0$

$$\lim_{t \rightarrow \infty} v_{o,ZS}(t) = \lim_{s \rightarrow 0} s \cdot \frac{2}{s + 1/RC} = 0$$

The results agree with the results of the initial and final value theorems.

$$V_o(s) = \underbrace{\frac{2}{s + 1/RC}}_{\text{zero-state}} - \underbrace{\frac{1}{s + 1/RC}}_{\text{zero-input}}$$

$$v_o(t) = \underbrace{2e^{-t/RC}u(t)}_{\text{zero-state}} - \underbrace{e^{-t/RC}u(t)}_{\text{zero-input}}$$

Initial values: $v_{o,ZI}(0_+) = \lim_{s \rightarrow \infty} s \cdot \frac{-1}{s + 1/RC} = -1$

$$v_{o,ZS}(0_+) = \lim_{s \rightarrow \infty} s \cdot \frac{2}{s + 1/RC} = 2$$

Final values: $\lim_{t \rightarrow \infty} v_{o,ZI}(t) = \lim_{s \rightarrow 0} s \cdot \frac{-1}{s + 1/RC} = 0$

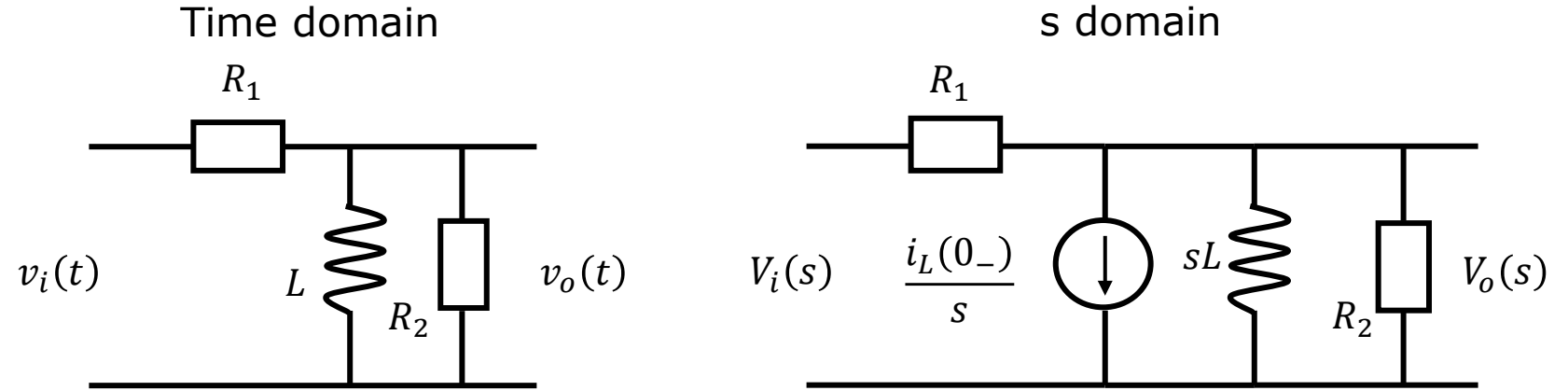
$$\lim_{t \rightarrow \infty} v_{o,ZS}(t) = \lim_{s \rightarrow 0} s \cdot \frac{2}{s + 1/RC} = 0$$

Question:

What kind of circuit is this?

Example 2.

Laplace transform
this circuit:



Here we only derive the input-output expression in the s-domain.

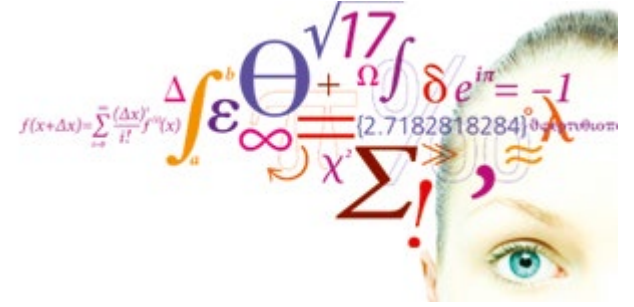
If we were asked to derive the response to a specific input signal, we would proceed with partial fraction expansion and inverse Laplace transform of the resulting terms.

$$\frac{V_i(s) - V_o(s)}{R_1} = \frac{V_o(s)}{sL} + \frac{i_L(0_-)}{s} + \frac{V_o(s)}{R_2}$$

The parallel model gives us a simple term to add.

$$V_o(s) = \frac{sR_1R_2L}{s(R_1 + R_2)L + R_1R_2} \cdot \frac{V_i(s)}{R_1} - \frac{R_1R_2L \cdot i_L(0_-)}{s(R_1 + R_2)L + R_1R_2}$$

$$V_o(s) = \underbrace{\frac{1}{R_1} \cdot \frac{sR_p}{s + R_p/L}}_{\text{zero-state}} \cdot V_i(s) - \underbrace{\frac{R_p \cdot i_L(0_-)}{s + \frac{R_p}{L}}}_{\text{zero-input}}; \quad R_p \stackrel{\text{def}}{=} \frac{R_1R_2}{R_1 + R_2}$$



Not in Lathi

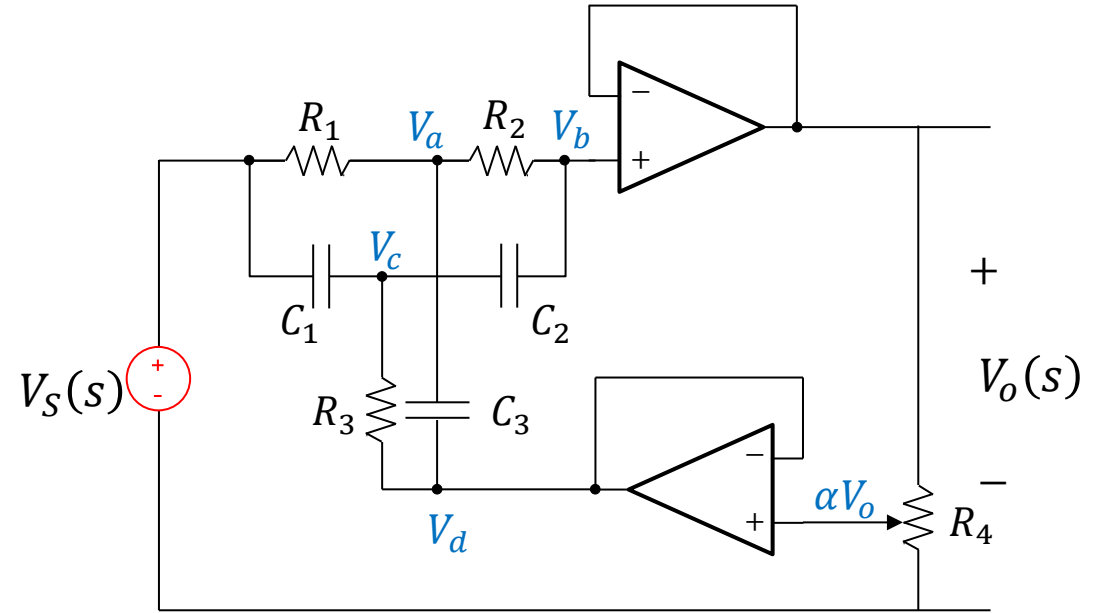
Another example, Notch filter

Deriving transfer function using the **current superposition method**.

The left-hand side expresses the partial current leaving the node, when all neighboring nodes are 0V. Because all neighboring nodes are at the same voltage, all components to the node are virtually in parallel and their parallel impedance is called the **shunt impedance**.

The right-hand side expresses the partial current entering the node, when the node voltage is zero.

We intend to use a CAS tool to solve the equations, hence we leave them in their most basic form, without inserting constraints.



$$a: \frac{V_a}{Z_a} = \frac{V_S}{R_1} + \frac{V_b}{R_2} + \frac{V_d}{Z_{C_3}}$$

$$\frac{1}{Z_a} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{Z_{C_3}}$$

$$b: \frac{V_b}{Z_b} = \frac{V_a}{R_2} + \frac{V_c}{Z_{C_2}}$$

$$\frac{1}{Z_b} = \frac{1}{R_2} + \frac{1}{Z_{C_2}}$$

$$c: \frac{V_c}{Z_c} = \frac{V_S}{Z_{C_1}} + \frac{V_b}{Z_{C_2}} + \frac{V_d}{R_3}$$

$$\frac{1}{Z_c} = \frac{1}{Z_{C_1}} + \frac{1}{Z_{C_2}} + \frac{1}{R_3}$$

Constraints: $V_b = V_o$ $V_d = \alpha V_o$ $Z_{C_x} = \frac{1}{sC_x}$

Solving for the transfer function:

$$H(s) = \frac{V_o(s)}{V_s(s)}$$

$$H(s) = \frac{C_1 C_2 C_3 R_1 R_2 R_3 s^3 + ((R_1 + R_2) R_3 C_1 C_2) s^2 + R_3 (C_1 + C_2) s + 1}{C_1 C_2 C_3 R_1 R_2 R_3 s^3 + ((R_1 + R_2) R_3 C_1 C_2 + (1 - \alpha) R_1 R_3 (C_1 + C_2) C_3 + (1 - \alpha) R_1 R_2 C_2 C_3) s^2 + (R_3 (C_1 + C_2) + (1 - \alpha) (R_1 + R_2) C_2 + (1 - \alpha) R_1 C_3) s + 1}$$

Motivated to simplify the design of the Twin-T notch filter, we will make some constraints on the component values:

$$R_1 = R_2 = 2R_3 = 2R$$

$$C_1 = C_2 = \frac{C_3}{2} = \frac{C}{2}$$

This reduces the expression significantly:

$$H(s) = \frac{(RC)^3 s^3 + (RC)^2 s^2 + (RC) s + 1}{(RC)^3 s^3 + (5 - 4\alpha)(RC)^2 s^2 + (5 - 4\alpha)(RC) s + 1}$$

We can seek common factors that cancel out:

$$H(s) = \frac{(RCs + 1)((RC)^2 s^2 + 1)}{(RCs + 1)((RC)^2 s^2 + 4(1 - \alpha)(RC) s + 1)}$$

We have obtained the transfer function of a **band-stop** filter.

It will have a notch at ω_0 .

A notch filter is required to have a narrow notch. The narrowness of the notch can be controlled by a parameter Q referred to as the **Quality factor** (DK: godhedsfaktor).

The parameter α represents a voltage divider in the form of a potentiometer (R_4). Hence the Q factor can be adjusted by turning R_4 .

$$H_{BS}(s) = \frac{((RC)^2 s^2 + 1)}{((RC)^2 s^2 + 4(1 - \alpha)(RC)s + 1)} \quad \omega_0 \stackrel{\text{def}}{=} \frac{1}{RC}$$

$$H_{BS}(j\omega) = \frac{\left(\frac{j\omega}{\omega_0}\right)^2 + 1}{\left(\frac{j\omega}{\omega_0}\right)^2 + \left(\frac{j\omega}{\omega_0}\right) 4(1 - \alpha) + 1}$$

$$H_{BS}(j\omega) = \frac{\left(\frac{j\omega}{\omega_0}\right)^2 + 1}{\left(\frac{j\omega}{\omega_0}\right)^2 + \frac{j\omega}{\omega_0} \frac{1}{Q} + 1}$$

$$Q = \frac{1}{4(1 - \alpha)} \Rightarrow \alpha = 1 - \frac{1}{4Q}$$

Full feedback: $\alpha = 1 \Rightarrow Q = \infty$

0 feedback: $\alpha = 0 \Rightarrow Q = 0.25$ $Q = 10 \Rightarrow \alpha = 0.975$

Twin-T notch filter

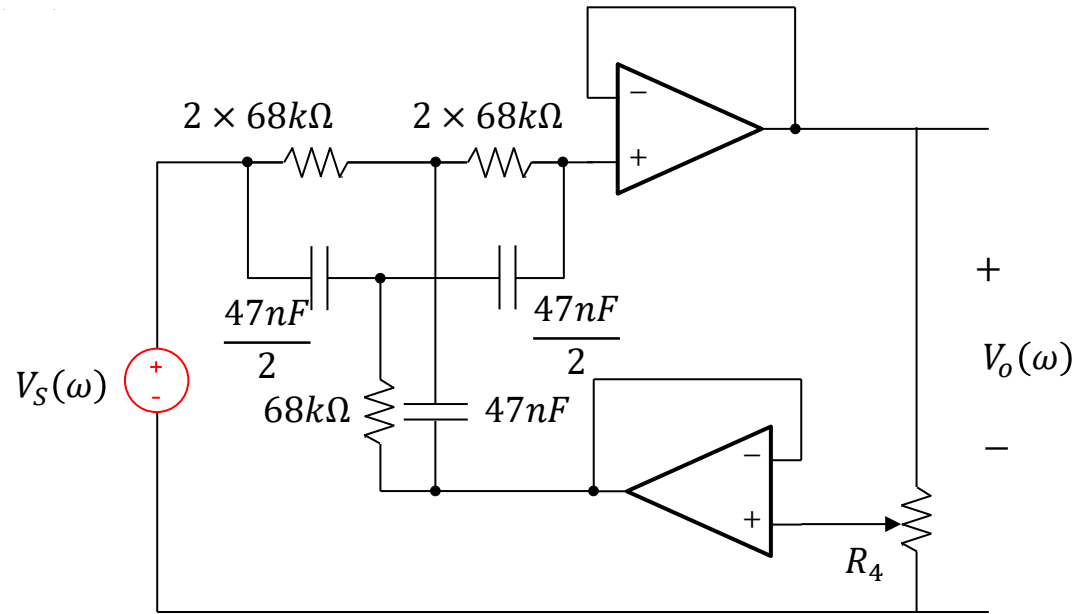
E3	E6	E12	E24
10	10	10	10
			11
		12	12
			13
	15	15	15
			16
		18	18
			20
22	22	22	22
			24
		27	27
			30
	33	33	33
			36
		39	39
			43
47	47	47	47
			51
		56	56
			62
	68	68	68
			75
		82	82
			91

$$\omega_0 = \frac{1}{RC} \quad R = \frac{1}{2\pi 50 \times C}$$

C	R	R1	R2	R3	C1	C2	C3
nF	kOhm	kOhm	kOhm	kOhm	nF	nF	nF
10,00	318,3	636,6	636,6	318,3	5,0	5,0	10,0
12,00	265,3	530,5	530,5	265,3	6,0	6,0	12,0
15,00	212,2	424,4	424,4	212,2	7,5	7,5	15,0
18,00	176,8	353,7	353,7	176,8	9,0	9,0	18,0
22,00	144,7	289,4	289,4	144,7	11,0	11,0	22,0
27,00	117,9	235,8	235,8	117,9	13,5	13,5	27,0
33,00	96,5	192,9	192,9	96,5	16,5	16,5	33,0
39,00	81,6	163,2	163,2	81,6	19,5	19,5	39,0
47,00	67,7	135,5	135,5	67,7	23,5	23,5	47,0
56,00	56,8	113,7	113,7	56,8	28,0	28,0	56,0
68,00	46,8	93,6	93,6	46,8	34,0	34,0	68,0
82,00	38,8	77,6	77,6	38,8	41,0	41,0	82,0

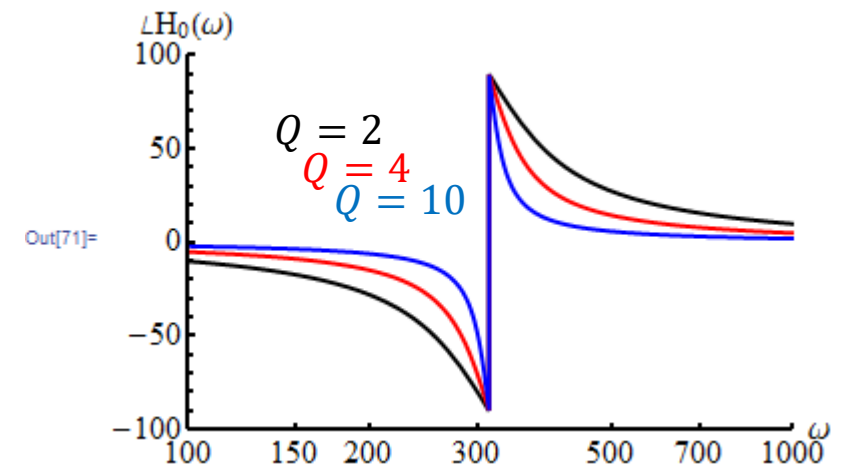
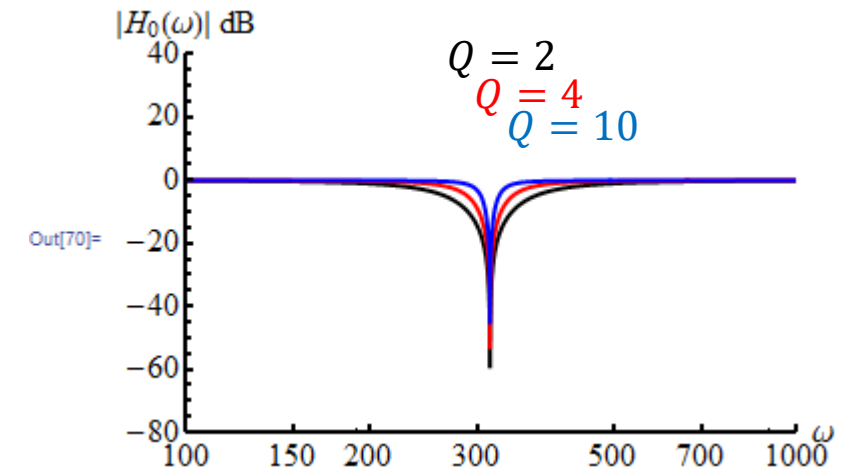
Choosing: $C = 47nF \Rightarrow R = 68k\Omega$

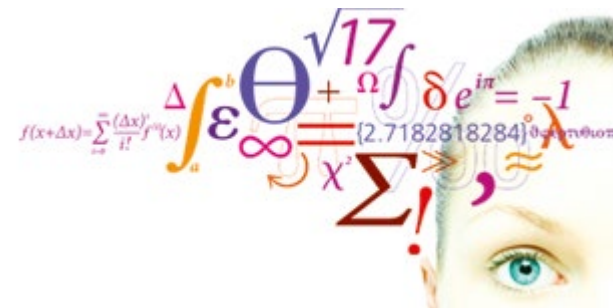
Twin-T notch filter



The Twin-T notch filter is a decent notch filter. However, placing the notch precisely at the desired frequency requires very precise component values, and the narrower the notch, the more precise component values need to be. An attenuation in the range 30 – 40 dB is a realistic goal.

We will return to this filter later and then investigate the relationship between the pole/zero positions and the behavior of the amplitude and phase spectra.





Problems

Filter 12: Lowpass filter – Question 1

Question 1.

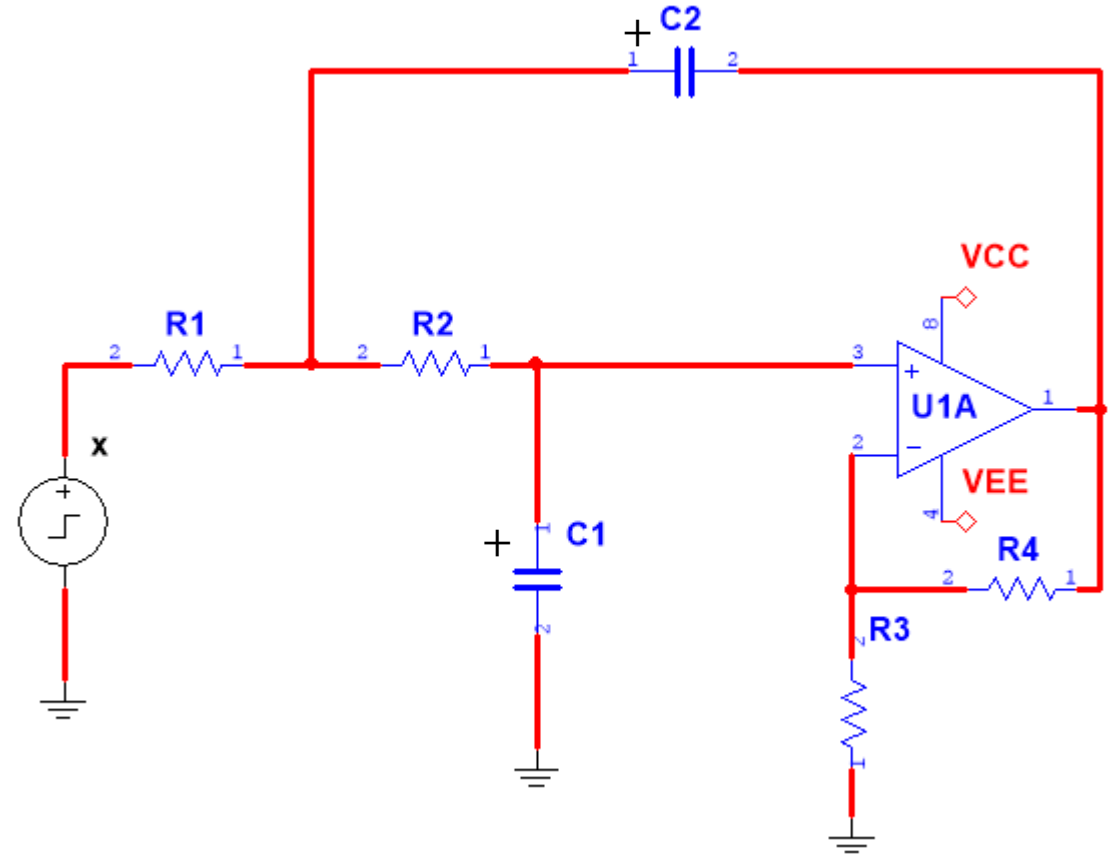
The circuit diagram is in a form suitable for time-domain analysis. We want to redraw the circuit, so it is suitable for s-domain analysis.

Laplace transform the frequency dependent components and the time-varying functions.

Include the initial value generators for the filter capacitors.

$$v_{C_1}(0_-) = 1V, v_{C_2}(0_-) = -1V.$$

A + sign indicates the positive side of the capacitor with respect to their voltage at $t = 0_-$. It is not an electrolytic capacitor.



Question 2.

Derive the system response:
$$Y(s) = \frac{P(s)}{Q(s)}X(s) + \frac{I(s)}{Q(s)}$$

Set up nodal equations and eliminate internal variables.

Answer:
$$Y(s) = \frac{b_0}{s^2 + a_1s + a_0}X(s) + \frac{\frac{K}{R_2C_1}v_{C_2}(0_-) + K\left(s + \frac{1}{R_2C_2} + \frac{1}{R_1C_2}\right)v_{C_1}(0_-)}{s^2 + a_1s + a_0}$$

Question 3

Solve for the zero-input response using partial fraction expansion and Laplace transform. Use Maple/Python.

Question 4

Solve for the zero-state response to the input $x(t) = 2u(t)$, using partial fraction expansion and Laplace transform. Use Maple/Python.

Answer: $y_{zs}(t) = (4 + 0.23 e^{-2680.2 t} - 4.23 e^{-147.25 t})u(t)$

Filter 13: Highpass filter - Question 1

Laplace transform the differential equation for the Sallen-Key highpass filter and derive $H(s)$. Look up the solutions to the problems on the first day of the course.

$$\ddot{y}(t) + \left(\frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} + \frac{1-K}{R_1 C_1} \right) \dot{y}(t) + \frac{1}{R_1 R_2 C_1 C_2} y = K \ddot{x}(t)$$

Answer:
$$H(s) = \frac{b_2 s^2}{s^2 + a_1 s + a_0}$$

$$H(s) = \frac{b_2 s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Filter 13: Highpass filter - Question 2

Calculate the impulse response $h(t)$ by inverse Laplace transformation of $H(s)$.

We remember that this filter is supposed to have a double root. Due to numerical rounding errors, Maple/Python may not produce a result, that indicates this.

Set the following parameter values and make some common sense rewriting of the results.

Adjusted coefficients

$$f0 := \frac{1}{20} : \omega_n := 2 \cdot \pi \cdot f0 : \\ \zeta := 1 :$$

$$\zeta = 1 \Rightarrow H(s) = \frac{b_2 s^2}{(s + \omega_n)^2}$$

Use this form for the inverse Laplace transform

Answer: $h(t) = 2\delta(t) - 0.6283(2 - 0.31415 t)e^{-0.31415 t}u(t)$

$$h(t) = 2\delta(t) - \frac{2\pi}{10} \left(2 - \frac{\pi}{10} t \right) e^{-\frac{\pi}{10} t}$$

Using precise coefficients

Calculate the zero-state response to $x(t) = 2 u(t)$ by calculating the inverse Laplace transform of $Y(s) = H(s)X(s)$.

Answer: $y_{zs}(t) = 4(1 - 0.31415 t)e^{-0.31415 t} u(t)$

Filter 14: Bandpass filter - Question 1

Laplace transform the differential equation for the Sallen-Key bandpass filter and derive $H(s)$. Look up the solutions to the problems on the first day of the course.

$$\ddot{y} + \left(\frac{1}{R_1 C_1} + \frac{1}{R_3 C_2} + \frac{1}{R_3 C_1} + \frac{1-K}{R_2 C_1} \right) \dot{y} + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \frac{1}{R_3 C_1 C_2} y = K \frac{\dot{x}}{R_1 C_1}$$

Answer:
$$H(s) = \frac{b_1 s}{s^2 + a_1 s + a_0}$$

Filter 14: Bandpass filter - Question 2

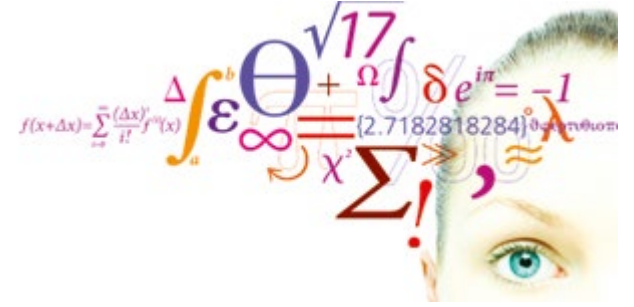
Calculate the impulse response $h(t)$ by inverse Laplace transformation of $H(s)$.

Answer:
$$h(t) = 629.1e^{-15.7t} \cos(314t + 0.05) u(t)$$

$$h(t) = 628.3 e^{-15.7t} \cos(314t) u(t) - 31.4 e^{-15.7t} \sin(314t) u(t)$$

Calculate the zero-state response to $x(t) = 2 u(t)$ by calculating the inverse Laplace transform of $Y(s) = H(s)X(s)$.

Answer:
$$y_{zs}(t) = 4e^{-15.7 t} \sin(313.76 t) u(t) = 4e^{-15.7 t} \cos\left(313.76 t + \frac{\pi}{2}\right) u(t)$$



Solutions

Filter 12: Lowpass filter – Question 1

Question 1.

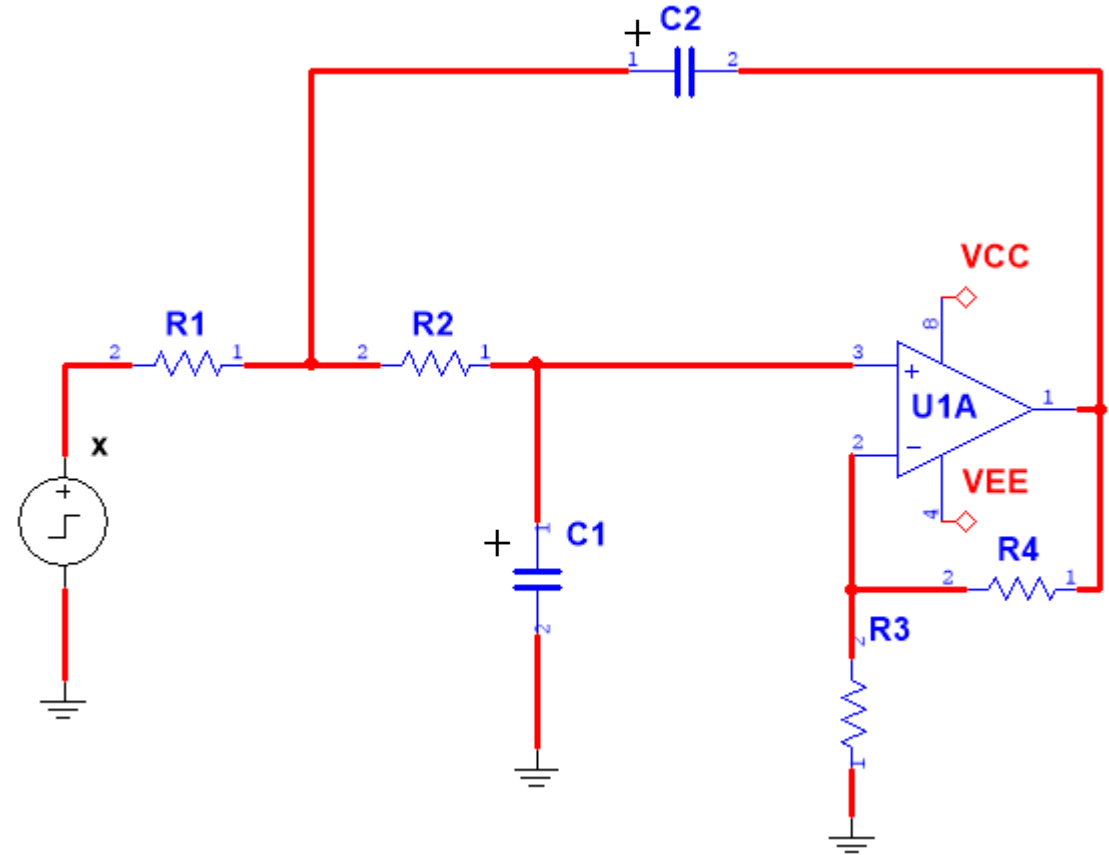
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Laplace transform the frequency dependent components and the time-varying functions.

Include the initial value generators for the filter capacitors.

$$v_{C_1}(0_-) = 1V, v_{C_2}(0_-) = -1V.$$

A + sign indicates the positive side of the capacitor with respect to their voltage at $t = 0_-$. It is not an electrolytic capacitor.



Filter 12: Lowpass filter – Question 1 (sol)

Question 1.

The circuit diagram is in a form suitable for time-domain analysis. We want to redraw the circuit, so it is suitable for s-domain analysis.

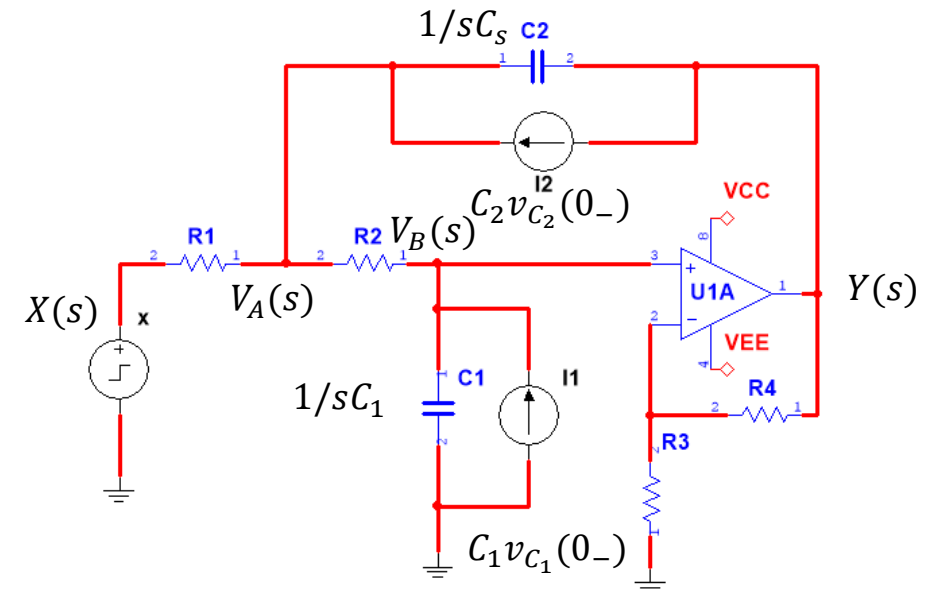
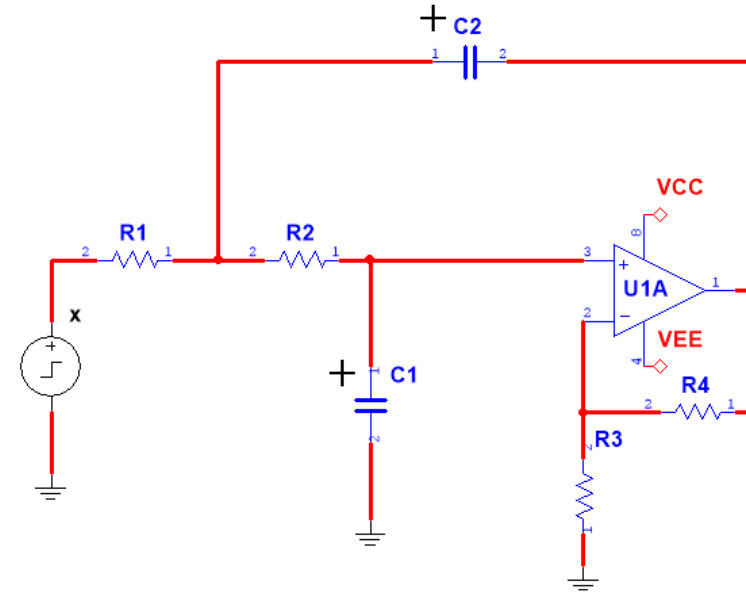
Laplace transform the frequency dependent components and the time-varying functions.

Include the initial value generators for the filter capacitors.

$$V_{C_1} = 1V, V_{C_2} = -1V.$$

I have chosen the **parallel component model**, because it is a sum of currents. This is most practical when we are writing nodal equations, where we sum up currents.

Series models are more convenient when we are using mesh analysis.



Filter 12: Lowpass filter – Question 2

Question 2.

Derive the system response:
$$Y(s) = \frac{P(s)}{Q(s)}X(s) + \frac{I(s)}{Q(s)}$$

Set up nodal equations and eliminate internal variables.

Answer:

$$Y(s) = \underbrace{\frac{b_0}{s^2 + a_1s + a_0}X(s)}_{\text{zero-state}} + \underbrace{\frac{\frac{K}{R_2C_1}v_{C_2}(0_-) + K\left(s + \frac{1}{R_2C_2} + \frac{1}{R_1C_2}\right)v_{C_1}(0_-)}{s^2 + a_1s + a_0}}_{\text{zero-input}}$$

Filter 12: Lowpass filter – Question 2 (sol)

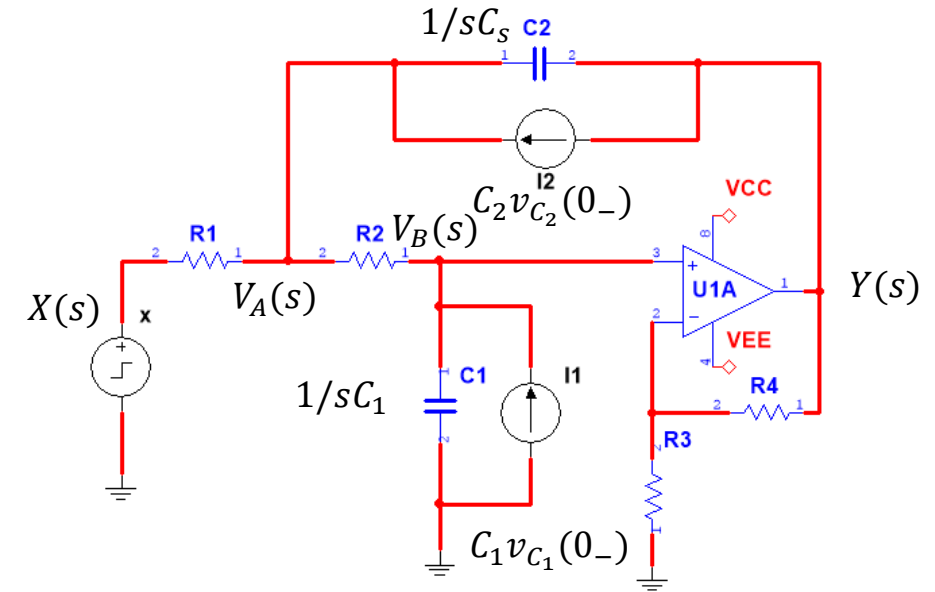
$$A1: -\frac{X(s) - V_A(s)}{R_1} + s C_2 (V_A(s) - Y(s)) - C_2 v_{C_2}(0_-) + \frac{V_A(s) - V_B(s)}{R_2} = 0$$

$$B1: -\frac{V_A(s) - V_B(s)}{R_2} + s C_1 V_B(s) - C_1 v_{C_1}(0_-) = 0$$

Constraint:

$$V_B(s) = \frac{1}{K} Y(s)$$

$V_B(s)$ is not an unknown. We should not solve for $V_B(s)$.



Filter 12: Lowpass filter – Question 2 (sol)

Setting up the nodal equations, remembering to add the initial conditions:

$$A1: -\frac{X(s) - V_A(s)}{R_1} + s C_2 (V_A(s) - Y(s)) - C_2 v_{C_2}(0_-) + \frac{V_A(s) - V_B(s)}{R_2} = 0$$

$$B1: -\frac{V_A(s) - V_B(s)}{R_2} + s C_1 V_B(s) - C_1 v_{C_1}(0_-) = 0$$

Insert the impedances in the Laplace domain:

Constraints

$$V_B := \frac{1}{K} \cdot Y$$

Equations

$$eqA1 := -\frac{(X - V_A)}{R1} + \frac{(V_A - Y)}{ZC2} - C2 v_{C20} + \frac{(V_A - V_B)}{R2} = 0$$

$$eqB1 := -\frac{(V_A - V_B)}{R2} + \frac{V_B}{ZC1} - C1 v_{C10} = 0$$

Solve

`solutions := simplify(solve({eqA1, eqB1}, [VA, Y], symbolic = true)) :`
`assign(solutions)`
`simplify(Y)`

$$-\frac{K(((C1 v_{C10} + C2 v_{C20}) R1 + C1 v_{C10} R2 + X) ZC2 + C1 R1 R2 v_{C10}) ZC1}{(-R1 - R2 - ZC1) ZC2 + R1((K - 1) ZC1 - R2)}$$

Impedances

$$ZC1 := \frac{1}{s \cdot C1} \quad ; \quad ZC2 := \frac{1}{s \cdot C2}$$

`simplify(Y)`

$$-\frac{((v_{C10} (C2 R2 s + 1) C1 + C2 v_{C20}) R1 + C1 v_{C10} R2 + X) K}{s((-C2 R2 s - 1) C1 + C2(K - 1)) R1 - R2 s C1 - 1}$$

Filter 12: Lowpass filter – Question 2 (sol)

Using various Maple tools to manipulate the expression:

`numer()`

`denom()`

`sort()`

`collect()`

`simplify(Y)`

$$- \frac{((vC10(C2R2s + 1)C1 + C2vC20)R1 + C1vC10R2 + X)K}{((-C2R2s - 1)C1 + C2(K - 1))sR1 - R2sC1 - 1}$$

`expl := numer(Y)`
`denom(Y)`

$$expl := \frac{K(C1R1R2vC10sC2 + C1R1vC10 + C1vC10R2 + C2R1vC20 + X)}{C1C2R1R2s^2 - C2KR1s + R1sC1 + R2sC1 + R1sC2 + 1}$$

`num1 := sort(numer(expl), s)`

$$num1 := (C1R1R2vC10C2s + C1R1vC10 + C1vC10R2 + C2R1vC20 + X)K$$

`den1 := sort(collect(denom(expl), s))`

$$den1 := C1C2R1R2s^2 + (-C2KR1 + C1R1 + C1R2 + C2R1)s + 1$$

Zero-state response:

$$Y_{zs} := s \rightarrow \frac{K}{R1 \cdot R2 \cdot C1 \cdot C2 \cdot s^2 + (R1 \cdot C1 + R2 \cdot C1 + (1 - K) \cdot R1 \cdot C2) \cdot s + 1} \cdot X:$$

Zero-input response:

$$Y_{zi} := s \rightarrow \frac{C1 \cdot R1 \cdot R2 \cdot vC10 \cdot C2 \cdot s + C1 \cdot R1 \cdot vC10 + C1 \cdot vC10 \cdot R2 + C2 \cdot R1 \cdot vC20}{R1 \cdot R2 \cdot C1 \cdot C2 \cdot s^2 + (R1 \cdot C1 + R2 \cdot C1 + (1 - K) \cdot R1 \cdot C2) \cdot s + 1} \cdot K:$$

$$Y_{zi2} := s \rightarrow \frac{K \cdot \left(s + \frac{1}{R2 \cdot C2} + \frac{1}{R1 \cdot C2} \right) \cdot vC10 + \frac{K}{R2 \cdot C1} \cdot vC20}{s^2 + \left(\frac{1}{R2 \cdot C2} + \frac{1}{R1 \cdot C2} + \frac{(1 - K)}{R2 \cdot C1} \right) \cdot s + \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}}:$$

Question 3

Solve for the zero-input response using partial fraction expansion and Laplace transform. Use Maple.

Solve for the zero-input response using partial fraction expansion and Laplace transform. Use Maple.

Zero input response

$$\text{convert}\left(\frac{K \cdot \left(s + \frac{1}{R2 \cdot C2} + \frac{1}{R1 \cdot C2}\right) \cdot vC10 + \frac{K}{R2 \cdot C1} \cdot vC20}{s^2 + \left(\frac{1}{R2 \cdot C2} + \frac{1}{R1 \cdot C2} + \frac{(1-K)}{R2 \cdot C1}\right) \cdot s + \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}}, \text{parfrac}, s\right)$$

$$\frac{2.116313058}{s + 147.3061597} - \frac{0.1163130577}{s + 2680.231742}$$

$$\text{invlaplace}\left(\frac{2.116313058}{s + 147.3061597} - \frac{0.1163130577}{s + 2680.231742}, s, t\right)$$

$$2.116313058 e^{-147.3061597 t} - 0.1163130577 e^{-2680.231742 t}$$

Question 4

Solve for the zero-state response to the input $x(t) = 2u(t)$, using partial fraction expansion and Laplace transform. Use Maple.

Answer: $y_{zs}(t) = (4 + 0.23 e^{-2680.2 t} - 4.23 e^{-147.25 t})u(t)$

Filter 12: Lowpass filter – Question 4 (sol)

Solve for the zero-state response to the input $x(t) = 2u(t)$, using partial fraction expansion and Laplace transform. Use Maple.

Answer: $y_{zs}(t) = (4 + 0.23 e^{-2680.2 t} - 4.23 e^{-147.25 t})u(t)$

Zero-state response

$$Y_{zs}(s) = H(s)X(s)$$

$$Y_{zs}(s) = H(s) \frac{2}{s}$$

$$\begin{aligned} & \text{convert} \left(\frac{\frac{K}{R1 \cdot R2 \cdot C1 \cdot C2}}{s^2 + \left(\frac{1}{R2 \cdot C2} + \frac{1}{R1 \cdot C2} + \frac{(1-K)}{R2 \cdot C1} \right) \cdot s + \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}}, \frac{2}{s}, \text{parfrac}, s \right) \\ & \quad \frac{4.000000000}{s} - \frac{4.232626116}{s + 147.3061597} + \frac{0.2326261165}{s + 2680.231742} \\ & \text{invlaplace} \left(\frac{4.000000000}{s} - \frac{4.232626116}{s + 147.3061597} + \frac{0.2326261165}{s + 2680.231742}, s, t \right) \\ & \quad 4. - 4.232626116 e^{-147.3061597 t} + 0.2326261165 e^{-2680.231742 t} \end{aligned}$$

Filter 13: Highpass filter - Question 1

Laplace transform the differential equation for the Sallen-Key highpass filter and derive $H(s)$. Look up the solutions to the problems on the first day of the course.

$$\ddot{y}(t) + \left(\frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} + \frac{1-K}{R_1 C_1} \right) \dot{y}(t) + \frac{1}{R_1 R_2 C_1 C_2} y = K \ddot{x}(t)$$

Answer:

$$H(s) = \frac{b_2 s^2}{s^2 + a_1 s + a_0}$$

$$H(s) = \frac{b_2 s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Filter 13: Highpass filter - Question 1 (sol)

Laplace transform the differential equation for the Sallen-Key highpass filter and derive $H(s)$. Look up the solutions to the problems on the first day of the course.

We are only asked to find $H(s)$.
We do not need to include the initial values because they do not multiply with $H(s)$.

Answer:

$$\ddot{y}(t) + \left(\frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} + \frac{1-K}{R_1 C_1} \right) \dot{y}(t) + \frac{1}{R_1 R_2 C_1 C_2} y = K \ddot{x}(t)$$

$$\ddot{y} + a_1 \dot{y} + a_0 y = b_2 \ddot{x}$$

$$s^2 Y(s) + a_1 s Y(s) + a_0 Y(s) = b_2 s^2 X(s)$$

$$(s^2 + a_1 s + a_0) Y(s) = b_2 s^2 X(s)$$

$$H(s) = \frac{Y_{zs}(s)}{X(s)} = \frac{b_2 s^2}{s^2 + a_1 s + a_0}$$

$$H(s) = \frac{b_2 s^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

Filter 13: Highpass filter - Question 2

Calculate the impulse response $h(t)$ by inverse Laplace transformation of $H(s)$.

When $\zeta = 1$, $s^2 + 2\zeta\omega_n s + \omega_n^2 = (s + \omega_n)^2$

Only in this form will Maple give us an inverse Laplace transform of the form $t e^{-a t}$. This is the form we associate with a repeated root.

Answer: $h(t) = 2\delta(t) - 0.6283(2 - 0.31415 t)e^{-0.31415 t}u(t)$

$$h(t) = 2\delta(t) - \frac{2\pi}{10} \left(2 - \frac{\pi}{10} t \right) e^{-\frac{\pi}{10} t}$$

Using precise coefficients

We remember that this filter is supposed to have a double root. Due to numerical rounding errors, Maple will not produce a result, that indicates this.

Set the following parameter values and make some common sense rewriting of the results.

Adjusted coefficients

$$f0 := \frac{1}{20} : \omega_n := 2 \cdot \pi \cdot f0 : \\ \zeta := 1 :$$

$$\zeta = 1 \Rightarrow H(s) = \frac{b_2 s^2}{(s + \omega_n)^2}$$

Use this form for the inverse Laplace transform

Filter 13: Highpass filter - Question 2 (sol)

Calculate the impulse response $h(t)$ by inverse Laplace transformation of $H(s)$.

Answer: $h(t) = 2\delta(t) - 0.6283(2 - 0.31415 t)e^{-0.31415 t}u(t)$

We remember that this filter is supposed to have a double root. Due to numerical rounding errors, Maple will not produce a result, that indicates this.

Set the following parameter values and make some common sense rewriting of the results.

Adjusted coefficients

$$f0 := \frac{1}{20} : \omega_n := 2 \cdot \pi \cdot f0 :$$

$$\zeta := 1 :$$

$$\zeta = 1 \Rightarrow H(s) = \frac{b_2 s^2}{(s + \omega_n)^2}$$

Use this form for the inverse Laplace transform

Impulse response

$$\text{invlaplace}\left(\frac{K \cdot s^2}{(s + \omega_n)^2}, s, t\right)$$

$$2 \cdot \text{Dirac}(t) + 0.06283185308 e^{-0.3141592654 t} (3.141592654 t - 20.)$$

Calculate the zero-state response to $x(t) = 2 u(t)$ by calculating the inverse Laplace transform of $Y(s) = H(s)X(s)$.

Answer: $y_{zs}(t) = 4(1 - 0.31415 t)e^{-0.31415 t} u(t)$

Filter 13: Highpass filter - Question 3 (sol)

Calculate the zero-state response to $x(t) = 2 u(t)$ by calculating the inverse Laplace transform of $Y(s) = H(s)X(s)$.

Answer: $y_{zs}(t) = 4(1 - 0.31415 t)e^{-0.31415 t} u(t)$

In this case it is better **not** to do a partial fraction expansion before the inverse Laplace transform.

Zero-state response

$$\text{invlaplace}\left(\frac{K \cdot s^2}{(s + \omega_n)^2} \cdot \frac{2}{s}, s, t\right)$$

$$-0.4000000000 e^{-0.3141592654 t} (3.141592654 t - 10.)$$

Filter 14: Bandpass filter - Question 1

Laplace transform the differential equation for the Sallen-Key bandpass filter and derive $H(s)$. Look up the solutions to the problems on the first day of the course.

$$\ddot{y} + \left(\frac{1}{R_1 C_1} + \frac{1}{R_3 C_2} + \frac{1}{R_3 C_1} + \frac{1-K}{R_2 C_1} \right) \dot{y} + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \frac{1}{R_3 C_1 C_2} y = K \frac{\dot{x}}{R_1 C_1}$$

Answer:
$$H(s) = \frac{b_1 s}{s^2 + a_1 s + a_0}$$

Filter 14: Bandpass filter - Question 1 (sol)

Laplace transform the differential equation for the Sallen-Key bandpass filter and derive $H(s)$. Look up the solutions to the problems on the first day of the course.

$$\ddot{y} + \left(\frac{1}{R_1 C_1} + \frac{1}{R_3 C_2} + \frac{1}{R_3 C_1} + \frac{1-K}{R_2 C_1} \right) \dot{y} + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \frac{1}{R_3 C_1 C_2} y = K \frac{\dot{x}}{R_1 C_1}$$

$$\ddot{y} + a_1 \dot{y} + a_0 y = b_1 \dot{x}$$

$$s^2 Y(s) + a_1 s Y(s) + a_0 Y(s) = b_1 s X(s)$$

Answer:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_1 s}{s^2 + a_1 s + a_0}$$

Calculate the impulse response $h(t)$ by inverse Laplace transformation of $H(s)$.

Answer:
$$h(t) = 629.1e^{-15.7 t} \cos(314 t + 0.05) u(t)$$

$$h(t) = 628.3 e^{-15.7 t} \cos(314 t) u(t) - 31.4 e^{-15.7 t} \sin(314 t) u(t)$$

Filter 14: Bandpass filter - Question 2 (sol)

Calculate the impulse response $h(t)$ by inverse Laplace transformation of $H(s)$.

Answer: $h(t) = 629.1e^{-15.7t} \cos(314t + 0.05) u(t)$

$$h(t) = 628.3 e^{-15.7t} \cos(314t) u(t) - 31.4 e^{-15.7t} \sin(314t) u(t)$$

Impulse repsonse

$$\begin{aligned} & \text{simplify} \left(\text{invlaplace} \left(\frac{b1 \cdot s}{s^2 + a1 \cdot s + a0}, s, t \right) \right) \\ & (628.3176032 \cos(313.7646688 t) \\ & \quad - 31.44542125 \sin(313.7646688 t)) e^{-15.70298545 t} \end{aligned}$$

Calculate the zero-state response to $x(t) = 2 u(t)$ by calculating the inverse Laplace transform of $Y(s) = H(s)X(s)$.

Answer:
$$y_{zs}(t) = 4e^{-15.7 t} \sin(313.76 t) u(t) = 4e^{-15.7 t} \cos\left(313.76 t + \frac{\pi}{2}\right) u(t)$$

Filter 14: Bandpass filter - Question 3 (sol)

Calculate the zero-state response to $x(t) = 2 u(t)$ by calculating the inverse Laplace transform of $Y(s) = H(s)X(s)$.

Answer: $y_{zs}(t) = 4e^{-15.7 t} \sin(313.76 t) u(t) = 4e^{-15.7 t} \cos\left(313.76 t + \frac{\pi}{2}\right) u(t)$

Zero-state response

$$\text{invlaplace}\left(\frac{b1 \cdot s}{s^2 + a1 \cdot s + a0} \cdot \frac{2}{s}, s, t\right)$$

$$4.005024565 e^{-15.70298545 t} \sin(313.7646688 t)$$

$$Y_{zs}(s) = H(s) \cdot X(s)$$

When we calculated the step response, we first Laplace transformed the step response: $X(s) = \frac{2}{s}$
Why didn't we do a similar thing when calculating the impulse response?

The Laplace transforms of $\delta(t)$ is $X(s) = 1$.

$$Y_{zs}(s) = H(s) \cdot X(s) = H(s) \cdot 1 = H(s)$$

$$y_{zs}(t) = \mathcal{L}^{-1}\{Y_{zs}(s)\} = \mathcal{L}^{-1}\{H(s)\} = h(t)$$

So, in an implicit way, we did do the same thing. We just didn't see it.